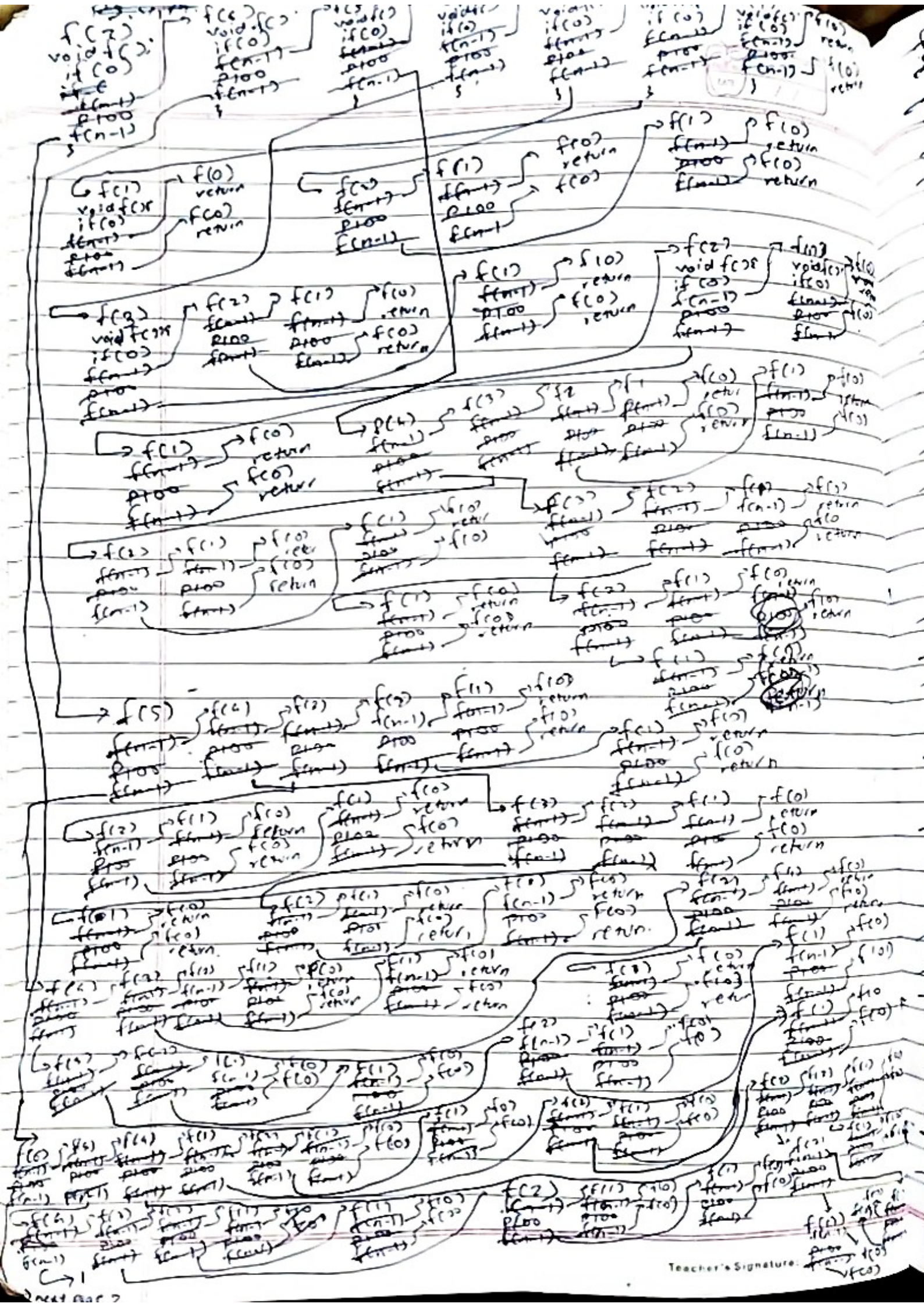

Teacher's Signature

3 2 1 0 1 2 3 4 X X Y Y X Y

Q. 1

```
void fun(int n){
  if(n==0){
    return;
  }
  s.o.p.(n);
  fun(n-1);
  s.o.p.(n+1);
}
```

~~f(0)~~
~~f(1)~~
~~f(2)~~
~~f(3)~~

call : fun(3);

n = 3

n=3

```
void fun(int n){
  if(n==0){
    return;
  }
  s.o.p.(n);
  fun(n-1);
  s.o.p(n+1);
}
```

n=2
~~s.o.p(n)~~
~~f(n-1)~~;
~~s.o.p(n+1)~~;

n=1
~~s.o.p(n)~~
~~f(n-1)~~
s.o.p(n+1)

n=0
if(n=0){
return
}

O/P: 3 2 1 2 3 4

Q. 2

```
void fun(int n){
  if(n==0){
    return;
  }
  s.o.p.("X");
  fun(n-1);
  s.o.p.("Y");
  f(n-1);
}
```

~~f(0)~~
~~f(0)~~
~~f(1)~~
~~f(0)~~
~~f(0)~~
~~f(1)~~
~~f(2)~~

call : fun(2);

n = 2

n=2
void fun(int n){
if(n==0){
return;
}
~~s.o.p(X)~~
~~fun(n-1)~~;
~~s.o.p(Y)~~
~~fun(n-1)~~
}

n=1
~~s.o.p(X)~~
~~fun(n-1)~~
~~s.o.p(Y)~~
~~fun(n-1)~~

n=0
return

n=0
return

n=1
~~s.o.p(X)~~
~~fun(n-1)~~
~~s.o.p(Y)~~
~~fun(n-1)~~

n=0
return

n=0
return

O/P:
X X Y Y X Y

Teacher's Signature.....

Q.

```
void f(int n){
  if(n <= 0){
    return ;
  }
  f(n-1);
  f(n-2);
}
```

Call:

f(4) n=4

→

n=4 → n=3 → n=2 → n=1 → n=0 → n=-1

~~f(n-1)~~ → ~~f(n-1)~~ → ~~f(n-1)~~ → ~~f(n-1)~~ → ~~f(n-1)~~ return → return

~~f(n-2)~~; ~~f(n-2)~~ ~~f(n-2)~~ ~~f(n-2)~~ ~~f(n-2)~~

↳ n=0 return

↳ n=1 → n=0 → n=-1

~~f(n-1)~~ return return

~~f(n-2)~~

↳ n=2 → n=1 → n=0 → n=-1

~~f(n-1)~~ → ~~f(n-1)~~ → return → return

~~f(n-2)~~ ~~f(n-2)~~

↳ n=0 return

Q.8

```
void fun(int n){
if (n==0){
  return;
}
f(n-1);
s.o.p.(n);
f(n-2);
}
```

call : f(4)
n = 4

f(0)				
f(1)				
f(2)				
f(3)				
f(4)				

→ n=4 → n=3 → n=2 → n=1 → n=0 → n=-1

f(n-1) → f(n-1) → f(n-1) → ~~f(n-1)~~ → return

s.o.p.(n) s.o.p.(n) s.o.p.(n) ~~s.o.p.(n)~~

f(n-2); f(n-2) f(n-2) f(n-2)

Infinite recursion

Q.9

```
void fun(int n){
if(n <= 1){
  s.o.p.(n);
  return;
}
f(n-1);
s.o.p.(n)
f(n-2)
```

call : f(3)
n = 3

~~f(0)~~	O/P:	0
~~f(1)~~	01-120301-1	1
~~f(0)~~		-1
~~f(-1)~~		2
~~f(0)~~		0
~~f(1)~~		3
~~f(2)~~		0
f(3)		1
		-1

n=3 → n=2 → n=1 → n=0 → n=-1 → n=0

~~f(n-1)~~ → f(n-1) → ~~f(n-1)~~ → ~~s.o.p.(n)~~ → ~~f(n-1)~~ → ~~s.o.p.(n)~~

~~s.o.p.(n)~~ ~~s.o.p.(n)~~ ~~s.o.p.(n)~~ return n<=1 s.o.p.(n) return return

~~f(n-2)~~ ~~f(n-2)~~ ~~f(n-2)~~

↳ n=1 → n=0 → n=-1

~~f(n-1)~~ ~~s.o.p.(n)~~ return ~~s.o.p.(n)~~ return

~~s.o.p.(n)~~

~~f(n-2)~~

Teacher's Signature:

Q. 6

```
void fun(int n)
if(n <= 1){
s.o.p.(n);
return;
}
s.o.p.(n);
f(n/2)
s.o.p.(n)
}
fun(8);
```

~~f(1)~~
~~f(2)~~
~~f(4)~~
~~f(8)~~

O/P :-

8
4
2
1
2
4
8

f(8) → f(4) → f(2) → f(1)

~~s.o.p.(n)~~ ~~f(n/2)~~ ~~s.o.p(n)~~

~~s.o.p(n)~~ ~~f(n/2)~~ ~~s.o.p(n)~~

~~s.o.p(n)~~ ~~s.o.p(n)~~ ~~return~~

Q 7

```
void fun(int n){
  if(n <= 0){
    return;
  }
  s.o.p.("A" + n);
  f(n-1)
  s.o.p.("B" + n);
  call f(3)
    n = 3
```

A 3
A 2
A 1
B 1
B 2
B 3

n = 3 → n = 2 → n = 1 → n = 0 return

~~s.o.p(A+n);~~ ~~s.o.p(A+n)~~ ~~s.o.p.(A+n)~~ ~~s.o.p(A+n)~~

~~f(n-1)~~ ~~f(n-1)~~ ~~f(n-1)~~ s.o.p.(B+n)

~~s.o.p(B+n);~~ ~~s.o.p.(B+n)~~ ~~s.o.p.(B+n)~~

Teacher's Signature:

0 2 0 4 0 2 0 | 5 3 1 1 3 5

1 4 1

Q3
```
void fun(int n){
    if(n <= 1){
        s.o.p.(n);
        return n;
    }
    f(n-2);
    s.o.p.(n)
    f(n-2);
}
call: f(4)
n = 4
```

n=4 → n=2 → n=0 → n=2 → n=0

f(n-2) → f(n-2) → s.o.p(n) → return → f(n-2) → f(n-2) → return

s.o.p(n) → s.o.p(n) → n=0 → f(n-2) → f(n=0) → s.o.p(n)

f(n-2) → f(n-2) → return

O/P:-

0 2 0 4 0 2 0

Q4
```
void fun(int n){
    if(n==0){
        return;
    }
    f(n-1)
    s.o.p(n*n);
    f(n-1)
}
call: f(2)
```

f(2) → f(1) → f(0) return

f(n-1) → f(n-1) → f(0) return

s.o.p(n*n) → s.o.p(n*n)

f(n-1) → f(n-1)

↳ f(1) → f(0) return

f(n-1) → f(0)

s.o.p(n*n)

f(n-1)

O/P: 1 4 1

Q5
```
void fun(int n){
    if(n<=0){
        return
    }
    s.o.p.(n)
    f(n-2)
    s.o.p(n)
}
call fun(5)
n=5
```

n=5 → n=3 → n=1 → n=-1

s.o.p(n) → s.o.p(n) → s.o.p(n) → n<=0

f(n-2) → f(n-2) → f(n-2) → return

s.o.p(n) → s.o.p(n) → s.o.p(n)

O/P: 5 3 1 1 3 5

Teacher's Signature: